

Data Cleaning and Preparation of an E-commerce Dataset

1. Introduction

Data cleaning is a critical step in data analysis. Raw datasets often contain missing values, duplicates, and inconsistencies. The goal of this project is to clean and prepare an e-commerce dataset by handling missing values, removing duplicates, correcting data formats, and validating data accuracy.

2. Dataset Overview

The dataset contains 1200 records and 14 columns including OrderID, Date, CustomerID, Product, Quantity, UnitPrice, TotalPrice, PaymentMethod, OrderStatus, and CouponCode.

3. Initial Data Inspection

The dataset was analyzed using Python. It was mostly structured with correct data types but required validation and cleaning.

4. Problems Identified

- Missing values in CouponCode (309 missing values)
- No major duplicate issues found
- Floating point precision caused false calculation mismatches

5. Data Cleaning Steps

- Missing values handled by replacing with 'No Coupon'
- Duplicate records removed
- Text standardized and cleaned
- Data types corrected and validated
- Calculation consistency checked using NumPy isclose

6. Data Analysis & Visualizations

Charts were created for product distribution, payment methods, and sales by order status to understand trends and patterns in the data.

7. Before vs After Cleaning

Missing values: 309 → 0

Duplicates: removed

Calculation errors: resolved using tolerance-based validation

8. Key Insights

The dataset was mostly clean but required validation. Product and payment trends provide useful business insights. Floating-point precision caused false mismatches which were resolved.

9. Tools Used

Python, Pandas, NumPy, Matplotlib

10. Conclusion

The dataset is now clean, consistent, and ready for analysis. All required cleaning tasks were successfully completed.